

Inside the Wave

State-Carrying Neural Pings, Delayed Observers, and Quantum-Like Inference

Antti (anttiluode)

2 October 2026 · Working paper v0.1

Abstract

Could an observer implemented by distributed, state-dependent neural waves experience a classical environment through a quantum-like geometry of inference? We formulate this question without identifying neuronal voltage, a transformer activation vector, a probability amplitude, and consciousness as the same object. A minimal observer contains history-dependent local states, waveform-bearing messages, propagation delays, and queries that alter the machinery they read. Its accessible present is assembled from a distribution of past events. In a narrowband approximation, delays become phase factors; coherent pathways can interfere; and sequential queries can fail to commute. With additional assumptions about linear mixing, quadratic detection, and outcome-conditioned state updates, parts of the resulting operational model admit a Hilbert-space description. We give explicit examples, a gauge-invariant loop-phase calculation, a projective response metric, and a classical stochastic-instrument formulation that exposes where those additional assumptions enter. We also prove a counterexample: delayed observation of a classical variable can remain ordinary Bayesian inference. Neither neural delay nor wave interference derives the Born rule as a fundamental law, a universal Planck constant, quantum entanglement, or Bell-inequality violations under Bell’s assumptions. Neural commitment to a report and suppression of alternative hypotheses resemble aspects of measurement phenomenology but do not establish quantum collapse or Everettian branching. The strongest defensible proposal is therefore a research program about the operational geometry of an embodied observer, with a more ambitious observer-origin account of physical quantum mechanics left as an explicitly incomplete conjecture. We specify inexpensive tests that could strengthen or narrow the observer model while keeping consciousness and fundamental physics outside the claims of those tests.

Status and provenance. This is an AI-assisted speculative mathematical working paper, developed from Antti’s state-ping and distributed-moment proposal. The equations below are proposed models and standard mathematical constructions, not a fitted biological theory. Analytic examples are identified as such. No new neuronal experiment, human study, quantum experiment, or ChessFly model run was performed for this paper. Source literature and the inspected repository are listed in Section 13.

1. The question: what is outside an observer made of waves?

The motivating picture is of a moment that is physically extended. Local neurons retain different histories; a message carries some signature of the state that produced it; receivers react according to their own state; and the resulting activity spreads with finite delays. There is no privileged miniature spectator at the center. What we call an observer is provisionally the evolving process that integrates these interactions and makes some of their consequences available for subsequent action and report.

The interesting question is then not merely whether the brain contains oscillations. It is whether the architecture of this process constrains the apparent properties of what it observes. A property might be available only through a particular query, at a particular phase, after a particular history. An internal description of the external world could therefore inherit features of the operations by which the observer obtains evidence.

This suggests three distinct hypotheses:

1. **Neural mechanism:** waveform features carry local history or state information that survives transmission and is useful to receivers.
2. **Observer geometry:** delayed, stateful querying creates interference, order effects, limited joint resolution, and context-dependent perceptual reports.
3. **Fundamental-physics conjecture:** physical quantum mechanics is entirely or partly the external-looking description generated by such an observer architecture.

The first does not entail the second, and the second does not entail the third. A mathematical model may support the second while providing a precise reason the third remains unexplained. This paper develops that middle territory and states the missing bridges.

The reference to consciousness is a further conditional assumption: *if* conscious experience is associated with the integrated activity, its experienced present could inherit the activity's temporal and operational structure. Nothing in the mathematical use of the word "observer" proves that this activity is conscious, identifies qualia, or excludes other accounts of consciousness.

2. What the biological starting point actually supports

Martin-Burgos and colleagues report state-dependent variability in intracellular action-potential waveforms [1]. Their controlled-input dataset contains two mouse cortical cells recorded at 200 kHz; their spontaneous-activity analysis includes 40 rat cortical neurons. Waveform features predict properties of injected drive and, in the simultaneous-recording dataset, aspects of peri-spike local field potential beyond a log-inter-spike-interval comparison.

The scale and heterogeneity matter. For local-field-potential amplitude, the reported population mean and median cross-validated R^2 are about 0.077 and 0.034; for its standard deviation, about 0.030 and 0.006. Population-level prediction was not significant for the spectral targets. These are selective statistical relationships, not a demonstration that one spike contains a complete network state. The supplied version was posted on 21 September 2026 and is a preprint not certified by peer review.

The study motivates preserving waveform information. It does not establish that the relevant somatic features survive axonal propagation unchanged, that a postsynaptic receiver decodes them, or that a globally coherent state can be reconstructed from them. Intracellular waveform variability also should not be identified directly with an extracellular travelling wave. These are different physical observables.

History dependence is compatible with ordinary membrane dynamics, including ionic conductances of the kind modeled by Hodgkin and Huxley [2]. Separately, recordings show travelling theta and alpha phase patterns in human neocortex [3]. Neither observation identifies a single brain-wide wave with consciousness. They make a stateful, distributed observer model worth formulating, while leaving its particular implementation open.

2.1 Carrying, recovering, and using state are separate tests

Let L denote a local history variable, M a waveform feature vector, and C measured covariates such as cell identity and inter-spike interval. A first encoding question is whether

$$I(L; M \mid C) > 0. \quad (1)$$

That is not yet a transmission or a usefulness claim. In a controlled Markov chain $L \rightarrow M \rightarrow Y$, conditional on fixed covariates, the data-processing inequality gives

$$I(L; Y \mid C) \leq I(L; M \mid C). \quad (2)$$

Here Y is what a receiver obtains through the tested channel. If the receiver also has correlated prior memory, that memory must be included explicitly; it cannot be credited to the message. A useful state ping requires recoverable information and a reader that uses it for a specified task. Information can be present in a coordinate and remain inaccessible under the chosen readout geometry.

3. A minimal delayed state-ping observer

3.1 Local dynamics, emitted waveforms, and reception

Let $w(t)$ be the hidden external state and $z_i(t)$ the local state of neural unit i , including whatever membrane variables and memory traces the model retains. The following is a generic model family, not a claim that a neuron literally implements each variable:

$$\dot{z}_i(t) = f_i(z_i(t), u_i(t)), \quad (3)$$

$$s_j(t) = \sum_k A_j(z_j(t_{jk}^-)) \kappa_j(t - t_{jk}; z_j(t_{jk}^-)), \quad (4)$$

$$u_i(t) = y_i(t) + \sum_j \int_0^\infty K_{ij}(v; z_i(t)) s_j(t - \tau_{ij} - v) dv. \quad (5)$$

t_{jk} are emission times, κ_j is a causal waveform, A_j controls its amplitude, $\tau_{ij} \geq 0$ is propagation delay, and K_{ij} is a causal receiving filter. State can alter waveform shape even when amplitude or spike count changes little. The kernels may have signed effects; a dynamical influence is not itself a probability.

The key interaction is asymmetric: the sender's history changes the emitted message, and the receiver's history changes what the message does. A ping need not contain a literal serialized state vector to carry a task-relevant signature of its origin. Nor need a physical neural query resemble attention's dot product.

With delays, the instantaneous vector $(z_1(t), \dots, z_N(t))$ is generally not a sufficient dynamical state. For bounded delays, an extended history segment such as

$$\mathcal{Z}_t = \{z_i(t + \theta) : -\tau_{\max} \leq \theta \leq 0\}_{i=1}^N \quad (6)$$

is needed, with filter states added or histories extended if the receiving kernels have longer memory. Calling one readout vector “the state” silently discards this distinction.

3.2 The accessible present is a retarded construction

A readout at lab time t can depend on a temporal distribution:

$$q(t) = \sum_i \int_0^\infty W_i(v) \phi_i(z_i(t - d_i - v)) dv. \quad (7)$$

Here d_i is the delay from a local source to the modeled integration process. q is a convenient description of accessible activity; it is not stipulated to be a quantum wavefunction or a conscious subject.

This is a thick present in an operational sense. Different contributions refer to different physical times. But finite transmission speed does **not** eliminate a global time coordinate or a spacetime foliation. A classical delayed network can be simulated on one clock. The observer lacks instantaneous access to every component, which is a constraint on access, not a theorem about the ontology of time.

The physically relevant comparison is dimensionless. For a component with angular frequency Ω , delay spread $\Delta\tau$ produces phase spread

$$\epsilon_{\text{phase}} = \Omega\Delta\tau. \quad (8)$$

A small absolute delay can matter at high frequency; a much larger delay can be negligible for a slowly changing variable. “Tiny time gradient” needs a frequency and a task before it becomes a quantitative statement.

3.3 The observer’s state is an equivalence class of usable histories

Let h be an available observation-and-action history and π an allowed future query policy. Define two histories to be operationally equivalent when

$$h \sim h' \iff P(R_{\text{future}} | h, \pi) = P(R_{\text{future}} | h', \pi) \text{ for every allowed } \pi. \quad (9)$$

R denotes observable reports or actions. The observer’s predictive state can be represented by the class $[h]$. This adapts the predictive-equivalence idea developed in computational mechanics [4] to a controlled query setting; it is not claimed as a new definition of state.

An important consequence follows. Which histories count as equivalent depends on what the observer can ask. Expanding its query repertoire can reveal distinctions it previously could not use. The experienced object’s “state” may therefore be query-relative even when the modeled external world has a definite classical state.

Equation (9) does not by itself produce a complex vector space, an inner product, or orthogonal subspaces. An operational quotient and a Hilbert space are different constructions. Their identification would require additional structure.

4. Delay alone does not force quantum probability

Consider a scalar classical world variable $x(t)$ observed through delayed channels:

$$y_i(t) = x(t - \tau_i) + \eta_i(t). \quad (10)$$

For nonnegative weights with $\sum_i a_i = 1$, a naive fused estimate is $\hat{x}(t) = \sum_i a_i y_i(t)$. A Taylor expansion gives, for a sufficiently smooth signal and short delays,

$$\text{bias } \hat{x}(t) \simeq -\bar{\tau}\dot{x}(t) + \frac{1}{2}\overline{\tau^2}\ddot{x}(t), \quad (11)$$

where the averages use the channel weights. Ignoring noise, the spread of the delayed samples around their weighted mean is approximately

$$\text{Var}_a[y_i(t)] \simeq \dot{x}(t)^2 \text{Var}_a[\tau_i]. \quad (12)$$

The observer can mistake temporal disagreement for external ambiguity. Yet the Bayesian posterior over $x(t)$ and its trajectory is an ordinary positive probability distribution. Its alternatives need no interference amplitudes.

Counterexample. Let $x(t) = x_0 + vt$, and suppose v and the delays are known. Then $y_i(t) + v\tau_i$ is a simultaneous estimate of $x(t)$. In the noiseless case every corrected channel agrees exactly, despite finite, unequal delays. With noise, standard likelihood fusion still applies.

More generally, a classically specified world, observer, and noise source admit a joint probability measure on their trajectories. Causal filtering and marginalization preserve that classical probability structure. A bounded observer may use a different, compressed representation for convenience, but delay alone does not compel it.

Thus asynchronous access can explain an apparent ambiguity, while leaving open whether a phase-sensitive observer has a more specific and interesting geometry.

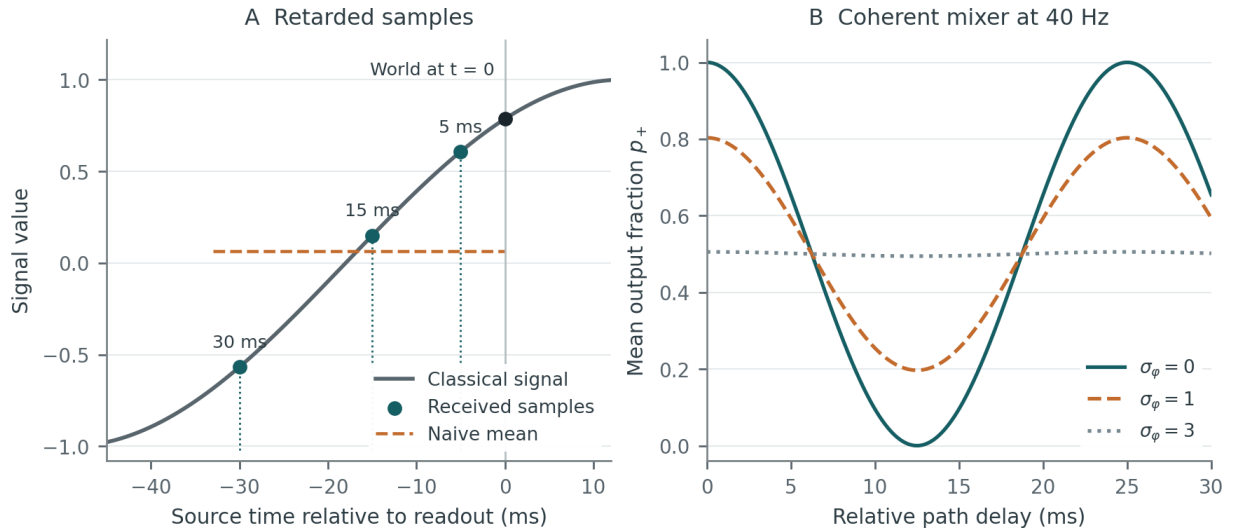


Figure 1: Analytic illustration. Left: three channels sample one classical sinusoid at different past times; their disagreement does not make the underlying signal a quantum superposition. Right: a different model coherently mixes two unit-energy paths at an illustrative carrier frequency of 40 Hz. Relative delay changes the output fraction, and Gaussian phase jitter reduces visibility according to Equation (17). All curves are specified examples, not neural measurements.

5. How delay can produce phase-sensitive inference

5.1 Complex amplitudes are a useful representation of classical waves

Suppose a population-level signal has an approximately narrowband component. Its real activity can be represented locally as $\text{Re}[a_i(t)e^{i\Omega t}]$, with a slowly varying complex envelope a_i . This is a signal representation; it does not postulate microscopic quantum coherence.

For a monochromatic component proportional to $e^{i\Omega t}$, a delay acts as

$$a_j(t - \tau_{ij}) = e^{-i\Omega\tau_{ij}} a_j(t). \quad (13)$$

For a slowly varying envelope this relation is approximate, with an envelope shift also present. A linearized coupling therefore contains a frequency-dependent factor

$$H_{ij}(\Omega) = g_{ij}e^{-i\Omega\tau_{ij}}, \quad (14)$$

before any additional phase of a receiving filter is included. Frequency-dependent transfer, damping, and nonlinearity remain part of the real network. Replacing them with one fixed unitary matrix would be an extra approximation.

This provides a concrete bridge from temporal geometry to amplitude geometry. Different histories and routes can arrive with different phases. A receiver that mixes them before a nonlinear readout can react to their relationship, rather than to either source's individual magnitude.

5.2 An explicit two-path example

Take two normalized inputs $a_1 = 1/\sqrt{2}$ and $a_2 = e^{i\varphi}/\sqrt{2}$. An ideal energy-preserving mixer produces

$$b_{\pm} = \frac{a_1 \pm a_2}{\sqrt{2}}, \quad |b_+|^2 + |b_-|^2 = 1. \quad (15)$$

The output fractions are

$$p_+ = \frac{1 + \cos \varphi}{2}, \quad p_- = \frac{1 - \cos \varphi}{2}. \quad (16)$$

These equations follow from coherent addition and quadratic intensity readout. At a carrier frequency $f = 40$ Hz, an illustrative relative delay of 12.5 ms gives $\varphi = \pi$ and $p_+ = 0$; zero relative delay gives $p_+ = 1$. The numbers describe this chosen mixer, not an observed neural phenomenon.

If independent Gaussian phase jitter has variance σ_φ^2 , averaging yields

$$\mathbb{E}[p_+] = \frac{1 + V \cos \bar{\varphi}}{2}, \quad V = e^{-\sigma_\varphi^2/2}. \quad (17)$$

Both the interference and its suppression are classical wave effects here. A single underlying classical trajectory can coexist with several coherent internal response pathways. Those pathways should not be called mutually exclusive world states unless a separate encoding model justifies that interpretation.

If two paths are read out separately before fusion, their intensities add and the cross term vanishes. The distinction is measurable: *where* the observer performs nonlinear detection changes the evidence it can use.

5.3 Local phase conventions and invariant loop phases

Under a change of local phase convention $a'_i = e^{i\chi_i}a_i$, the same coupling becomes

$$H'_{ij} = e^{i\chi_i}H_{ij}e^{-i\chi_j}. \quad (18)$$

For a directed closed route C , the product of its nonzero edge couplings has an invariant argument:

$$\Phi_C(\Omega) = \arg \prod_{(j \rightarrow i) \in C} H_{ij}(\Omega). \quad (19)$$

The local phases telescope. For positive real gains and pure delays,

$$\Phi_C(\Omega) = -\Omega \sum_{(j \rightarrow i) \in C} \tau_{ij} \pmod{2\pi}. \quad (20)$$

One can remove edge phases on a chosen spanning tree by changing local conventions, but cannot remove an arbitrary nonzero cycle phase everywhere. Filter phases or signed gains contribute further terms. This is a limited but exact sense in which the network has gauge-like phase structure.

A loop phase is not evidence of curved spacetime, electromagnetism's gauge field, or a quantum field. It is a classical signal-network invariant. Its possible computational role is more concrete: relative return phase can select which recurrent modes are reinforced. The meaningful research question is whether an observer uses such invariants to organize incoming evidence and memory.

5.4 The apparent geometry can be a geometry of responses

There is a stronger mathematical version of the idea that the observer's machinery shapes its apparent world. Suppose an external condition together with the observer's history has coordinates θ , and the wave encoder produces a smooth, nonzero response $a(\theta)$. Normalize it to $\psi(\theta)$. Restrict attention to readouts invariant under a common phase, and assume coherent comparison with a reference response is available. An ideal detector matched to $\psi(\theta)$ transmits the fraction

$$F(\theta, \theta + d\theta) = |\psi(\theta)^\dagger \psi(\theta + d\theta)|^2. \quad (20a)$$

Expanding $1 - F$ to second order gives

$$1 - F = g_{\mu\nu} d\theta^\mu d\theta^\nu + O(\|d\theta\|^3), \quad (20b)$$

$$g_{\mu\nu} = \text{Re} \left[(\partial_\mu \psi)^\dagger (\partial_\nu \psi) - (\partial_\mu \psi)^\dagger \psi \psi^\dagger (\partial_\nu \psi) \right]. \quad (20c)$$

This is the standard projective Hilbert-space metric pulled back through the response encoder [15]. The construction can be applied to the normalized classical waves used here. It measures mismatch between responses under the assumed coherent detector, rather than ordinary distance between objects. If the encoding loses distinctions, the pulled-back matrix has null directions; if only one fixed detector is available, even this geometry can overstate what the observer distinguishes.

For the two-path family $\psi(\varphi) = (1, e^{i\varphi})/\sqrt{2}$, the formula gives $g_{\varphi\varphi} = 1/4$. If $\varphi = \Omega\tau$, then $g_{\tau\tau} = \Omega^2/4$. The same small physical delay is more distinguishable at higher carrier frequency. Delays separated by a full carrier period produce the same response in this ideal family, unless some additional channel resolves the ambiguity.

Allowing relative amplitude as well as phase gives an even more recognizable geometry. For

$$\psi(\vartheta, \varphi) = \begin{pmatrix} \cos(\vartheta/2) \\ e^{i\varphi} \sin(\vartheta/2) \end{pmatrix}, \quad ds^2 = \frac{1}{4} (d\vartheta^2 + \sin^2 \vartheta d\varphi^2), \quad (20d)$$

the response metric is a sphere of radius $1/2$: the familiar pure two-level state geometry, realized here as normalized classical two-channel waves modulo common phase. The curved response space comes from the encoder and the equivalence of its readouts; it does not require a physically curved

external world. This is an explicit example of a quantum-like representational geometry arising inside the specified observer, with no claim that neuronal messages actually instantiate it.

From inside this restricted machinery, closeness could therefore mean “evokes nearly the same accessible response,” and a direction could mean “changes the response in a particular way.” Its apparent geometry depends on frequency, history, encoding, and available queries. This is a concrete route to a quantum-looking geometry of representations. It does not derive a Lorentzian spacetime metric, gravity, or the external material world’s laws. Establishing such a relation would require an independent map from external physical preparations to these response coordinates.

6. Queries that change what they query

6.1 A classical instrument description

Include the world, observer memory, and relevant delay histories in a classical state X . For a query A and outcome a , let J_a^A be a nonnegative transition kernel. The outcome kernels are subnormalized, and their sum is stochastic. For a row distribution μ ,

$$P(a \mid A, \mu) = \mu J_a^A \mathbf{1}, \quad \mu_{A,a} = \frac{\mu J_a^A}{\mu J_a^A \mathbf{1}}. \quad (21)$$

The same kernel describes the observation probability and the physical state change. Sequential outcomes have probabilities

$$P(a, b \mid A \text{ then } B, \mu) = \mu J_a^A J_b^B \mathbf{1}. \quad (22)$$

In general $J_a^A J_b^B \neq J_b^B J_a^A$. That is sufficient for order effects. Nothing in this construction requires negative probabilities or quantum states.

For an elementary counterexample, let query A reset a classical bit to zero and B flip it:

$$K_A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad K_B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (23)$$

With $\mu = (1, 0)$, $\mu K_A K_B = (0, 1)$ but $\mu K_B K_A = (1, 0)$. State-altering questions fail to commute in a perfectly classical two-state machine. Underlying classical observable functions still commute under multiplication; these are different intervention sequences, not merely two passive readings of one unchanged state.

6.2 A Hilbert-space description can also have order effects

For comparison, adopt normalized amplitudes ψ and orthogonal projectors P and Q , *including* quadratic outcome probabilities and projective conditional updating as model assumptions. Then

$$P(P=1, Q=1) = \|QP\psi\|^2, \quad P(Q=1, P=1) = \|PQ\psi\|^2. \quad (24)$$

For

$$\psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (25)$$

the two joint probabilities are $1/2$ and $1/4$. The complementary outcomes complete each measurement; these are not probabilities conditioned on both outcomes already having occurred. The same algebra describes intensities through ideal classical linear filters.

Quantum-probability models of human question-order effects already exist. Wang and colleagues tested a particular equality predicted by such a model [5]. That work provides a precedent for using quantum mathematics operationally in cognition, not evidence that neuronal tissue generates fundamental quantum mechanics. Our contribution is a proposed mechanistic route through delayed, history-dependent messages; it is not the discovery of noncommuting questions.

A rigorous comparison must ask whether a restricted amplitude model predicts held-out behavior better than equally constrained classical memory models. Order dependence by itself cannot distinguish them. Nor is everyday context dependence equivalent to a demonstration of quantum contextuality: the operational equivalences and disturbance controls of any such test must be specified.

7. What must be added to obtain a Born-like readout?

7.1 A conditional derivation from quadratic detection

Suppose $a \in \mathbb{C}^n$ represents internal waves, an orthonormal detector basis is $\{v_b\}$, and event rates satisfy

$$\lambda_b = \gamma |v_b^\dagger a|^2. \quad (26)$$

If the detector channels are independent Poisson processes with constant rates during a decision interval, the probability that channel b fires first is

$$P(b) = \frac{\lambda_b}{\sum_c \lambda_c} = \frac{|v_b^\dagger a|^2}{\|a\|^2} = |v_b^\dagger \psi|^2, \quad \psi = \frac{a}{\|a\|}. \quad (27)$$

The first equality is a classical race calculation. The second uses completeness of the orthonormal basis. The last has the Born form. This is a useful existence example: a classical wave-and-detector machine can implement a Born-shaped sampling rule for its internal channels.

The assumptions do the work. Linear coherent mixing, quadratic rates, common gain, a complete basis, and the Poisson race have not been derived from state-dependent spikes. A neural detector could instead use thresholds, rectified amplitudes, or a softmax of learned scores. A different transduction rule generally gives different probabilities.

For a positive-operator readout $E_b \succeq 0$ with $\sum_b E_b = I$, the same quadratic construction gives $P(b) = \psi^\dagger E_b \psi$. This enlarges the modeled readout family but does not show that all such readouts are biologically available, or derive an outcome-conditioned update rule. Matching one probability formula is weaker than matching quantum preparations, transformations, measurements, and their composition.

7.2 Why a state vector is not automatically a wavefunction

Any classical distribution can be embedded as $\psi_j = \sqrt{p_j} e^{i\theta_j}$. If all readouts are diagonal in that basis,

$$\psi^\dagger \text{diag}(g_j) \psi = \sum_j p_j g_j. \quad (28)$$

The phases do nothing. Allowing non-diagonal measurements is a substantive physical or computational assumption; it is not licensed by writing square roots.

Likewise, a transformer’s residual vector is a learned real coordinate system with architecture-dependent operations. An attention score or recurrent activation does not automatically possess complex phase, conserved norm, tensor-product composition, or a Born probability rule. Its useful role here is as an analogy for a distributed query, not an identification of mathematical objects.

7.3 Why writing a Schrodinger equation is not a derivation of physics

For a complex linear dynamical system $\dot{a} = Ma$, its Euclidean norm is conserved for every initial state exactly when

$$M^\dagger + M = 0. \quad (29)$$

Only in this special case can one set $H = i\hbar_{\text{eff}}M$ and obtain a Hermitian H with

$$i\hbar_{\text{eff}}\dot{a} = Ha. \quad (30)$$

For another positive inner-product metric G , the corresponding condition is $M^\dagger G + GM = 0$. Typical damped, driven, delayed neural dynamics do not satisfy these conservation conditions. A delay system also generally needs an extended history state, not one finite-dimensional generator.

$\hbar_{\text{eff}}$ here is an arbitrary conversion constant unless independently fixed by an energy-action relation. Equation (30) does not derive the measured Planck constant. It demonstrates a mathematical resemblance that must be kept separate from a physical explanation.

8. Complementarity, commitment, and the branching picture

8.1 Time-frequency resolution is a genuine wave constraint

For a normalized wave packet $g(t)$ with finite variances and angular-frequency Fourier transform

$$\tilde{g}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt, \quad (31)$$

the spreads defined by $|g(t)|^2$ and $|\tilde{g}(\omega)|^2$ obey

$$\Delta t \Delta \omega \geq \frac{1}{2}, \quad \Delta t \Delta f \geq \frac{1}{4\pi}. \quad (32)$$

This is the Fourier uncertainty relation used in communication theory, including Gabor’s analysis [6]. A Gaussian packet attains the bound. An observer using such packets has a resolution tradeoff between temporal localization and spectral concentration.

However, this is not a universal theorem about every estimator of every stimulus. Known signal structure, repeated observations, or additional channels can support more precise parameter estimation than the naive packet-width interpretation suggests. And ω is not automatically particle momentum or energy. Deriving $\Delta x \Delta p \geq \hbar/2$ for external matter requires the appropriate physical variables and action scale, not simply a neural time window.

8.2 Perceptual commitment can be modeled as an attractor transition

A simple noisy commitment variable is

$$d\xi = (\alpha\xi - \beta\xi^3 + h(t))dt + \sqrt{2D} dW_t, \quad \alpha, \beta > 0. \quad (33)$$

It moves in a double-well potential $V(\xi) = \beta\xi^4/4 - \alpha\xi^2/2 - h\xi$. Evidence biases the competing basins, and a stable sign can support a definite report. Prior uncertainty and a later categorical commitment can thus coexist in a classical dissipative process.

This can model one aspect of the experience of obtaining a result. It neither derives the Born weights nor solves the quantum measurement problem. Its transition rates depend on its potential, noise, and evidence. Physical energy expenditure is expected in biological processing, but an energetic cost alone does not specify outcome probabilities or turn an attractor into a projective measurement.

8.3 Counterfactual branches are not Everettian worlds

An observer can maintain several candidate futures and suppress all but one report. That is a meaningful computational branching picture. The candidates can even occupy different physical pathways in the same brain. But representational alternatives are not automatically distinct external worlds.

Everett's relative-state formulation starts from a quantum description of system and observer [7]. An unpruned collection of neuronal hypotheses lacks the required universal unitary dynamics and quantum composition merely by being unpruned. Neural inhibition changes or damps activity; it is not, by that fact, the entanglement-mediated suppression of interference described by quantum decoherence [8].

The useful analogy is narrower: the observer's available report may track one stable thread while more possibilities existed in its earlier dynamics. Explaining that computational selection would be valuable without claiming to explain the existence or nonexistence of many worlds.

9. The fundamental boundary: Bell correlations and public records

9.1 A delayed local classical observer remains Bell-local under the assumptions

Let two separated devices choose settings a, b and return outcomes $A, B \in \{-1, +1\}$. Include all their shared history and local memory in λ . Bell-local factorization and setting independence are

$$P(A, B | a, b, \lambda) = P(A | a, \lambda)P(B | b, \lambda), \quad \rho(\lambda | a, b) = \rho(\lambda). \quad (34)$$

For deterministic assignments, the CHSH combination is

$$S_\lambda = A_0(B_0 + B_1) + A_1(B_0 - B_1) = \pm 2. \quad (35)$$

The stochastic case follows by adding local randomness to λ and averaging. Thus $|S| \leq 2$ [9,10]. Delays and memory do not remove this constraint while the stated independence, locality, and trial-selection assumptions remain in force.

Loophole-free experiments have violated a Bell inequality; Hensen and colleagues reported a separated-electron-spin test that explicitly allowed device memory in its analysis [11]. A theory assigning the correlations solely to classical neural integration must therefore explain which assumption fails and reproduce the full experimental arrangement. "The observer has no instantaneous global now" is not such an explanation.

A central processor that sees both settings before constructing reports can manufacture different correlations, but it then violates the relevant locality condition. Setting-dependent selection or

coincidence filtering can similarly change a dataset. Neither maneuver is a derivation of the properly constrained experiment. This argument does not rule out every conceivable interpretation of quantum theory; it rules out the specified local classical replacement under the stated assumptions.

9.2 The observer-replacement test

Write an ordinary recording chain as

$$P_{\mathcal{A}}(r | u) = \sum_d C_{\mathcal{A}}(r | d) P_{\text{det}}(d | u), \quad (36)$$

where d is a detector record, u the experimental setting, and $C_{\mathcal{A}}$ the report channel of observer architecture $\mathcal{A}$. Changing a biological observer can change the final reports without changing the raw detector records. With faithful recording and appropriate calibration, independently implemented observers can recover the same P_{det} .

This is a practical dividing line. A theory of human temporal integration should predict architecture-dependent reporting effects. A replacement for physical quantum mechanics must additionally explain the detector statistics recorded by automated apparatus, including when no human watches a live display. Reassigning the word “observer” to the detector does not close this gap; the detector’s own physical laws must then be modeled and tested.

QBism treats quantum state assignments as an agent’s probability judgments while retaining quantum theory’s rules [12]. Relational quantum mechanics relates state descriptions to physical systems and does not require consciousness as a special ingredient [13]. These are relevant conceptual neighbors. Neither establishes that cortical delays generate those rules. The present proposal must not borrow their interpretive vocabulary as evidence for its neural mechanism.

10. What classicality could mean in this model

There are two questions: when an observer gives stable, approximately classical reports, and how classical descriptions of physical matter arise within quantum theory. A model of the first is not automatically a solution to the second.

Let T_{world} be the timescale of the external variable relevant to a task, T_{int} the integration window, T_{settle} the reporting relaxation time, and $\Delta\tau$ the arrival-delay spread. Stable tracking is plausible when, at adequate signal-to-noise ratio,

$$\frac{\Delta\tau}{T_{\text{world}}} \ll 1, \quad \frac{T_{\text{int}}}{T_{\text{world}}} \ll 1, \quad \frac{T_{\text{settle}}}{T_{\text{world}}} \ll 1. \quad (37)$$

Known delays can also be compensated by prediction, as Section 4 shows. Meanwhile phase averaging can remove cross terms. For the wave model one may define

$$\rho = \frac{\mathbb{E}[aa^\dagger]}{\mathbb{E}[\|a\|^2]}. \quad (38)$$

If relative phases average independently, relevant off-diagonal entries approach zero and quadratic output statistics become mixtures of modal intensities. This is classical loss of wave coherence in the specified model, not a derivation of environmental quantum decoherence.

These limits do not make macroscopic classicality a “high-energy limit.” Energy, delay, coherence, action, environmental coupling, and task timescale are different quantities. A high-frequency internal component can become more sensitive to a fixed delay. A slowly varying, low-energy

signal can be easy to track. The meaningful observer limit involves predictability, calibration, phase stability, and resolution.

What may emerge is a reliable **public present**: different observers predict and act on compatible records, even though their physical integrations are asynchronous. The strongest insight in the distributed-moment picture may be that simultaneity of access is unnecessary for agreement about a world.

11. Connection to ChessFlyStatePings and inexpensive tests

11.1 What the existing repository contributes

ChessFlyStatePings [14] is an artificial recurrent-network harness around the frozen ChessFly checkpoint and a fly-derived graph. It is not a biophysical fly brain. We inspected its main branch at commit 49385519e55cb7e3b2a06bf3f512ba6b3967bebc, including the committed query-memory summary.

Its initial fast-minus-slow residue was largely collinear with present activity. Subsequent gates projected that component away and compared the remaining direction against norm-matched controls. A three-position directional-specificity smoke test suggested stronger effects for some trained readouts than for others. That is receiver sensitivity, not demonstrated useful information or improved chess ability.

The literal Gate 2 memory-address experiment then gave the following committed result:

Readout arm	Top-1 retrieval	Mean reciprocal rank	Mean correct margin
Present coordinate	1.000	1.000	+0.006764
Present + real history	0.333	0.611	-0.006120
Present + shuffled history	1.000	1.000	+0.005393

The memory bank and equal-weight cosine scoring are important parts of that negative result. A coordinate unusually potent for a frozen decoder was not automatically a useful key to its own recorded predecessor. This is precisely why carrying, recovering, and using state must be tested separately.

These data neither prove nor disprove consciousness or an observer-origin theory of quantum mechanics. They constrain one much smaller message-and-reader construction. This paper does not rerun or modify ChessFlyStatePings; the experiment code and original receipts remain in that repository.

11.2 A proposed sequence of tests

The following tests are proposals. Apart from the existing ChessFly results just cited and the analytic calculations in this paper, they have **not** been executed.

Test A: state-bearing messages. Build a small classical world with two histories that end at the same present observation but imply different future trajectories. Compare timestamp-only messages with waveform features at matched communication budgets. Train a receiver on training trajectories only, and measure held-out predictive log loss on whole trajectories. Include shuffled history, randomized waveform shape with matched basic amplitude/duration, and a task-independent hidden-variable control. A failure narrows the claimed usefulness of the chosen waveform code.

Test B: delay-phase readout. Use a specified two-path oscillator and independently measured receiver nonlinearity. Sweep relative delay and carrier frequency; predict the cross term before examining held-out settings. Shuffle relative phase while preserving marginal powers. Compare mixing before quadratic detection against detecting each path separately before fusion. Equation (17) predicts a visibility curve only for the stated jitter model. Success demonstrates internal classical interference, not quantum physics.

Test C: query order and memory. Apply two defined interventions in both orders, starting from matched initial histories. Measure response differences and physical disturbance. Compare a restricted amplitude model against a classical stochastic-instrument model with comparable parameter budgets and the same training data. Score held-out sequences, not only the pairs used to fit the model. If the amplitude model offers no predictive or descriptive advantage, the quantum-like representation is optional for that task.

Test D: the slow and calibrated limits. Sweep world timescale, delay spread, integration window, phase jitter, and calibration quality independently. Compare the naive estimator with a delay-aware predictor. Declare in advance which readout becomes stable under which limit. A delay-aware classical estimator succeeding where an uncalibrated wave observer fails would support an access-and-inference explanation of the ambiguity.

Test E: observer replacement. Feed identical preserved raw records into observers with different internal architectures and compare both their calibrated output statistics and their immediate reports. This probes the distinction in Equation (36). It is not a Bell experiment. Any future attempt to claim new fundamental physics would need an independently specified physical experiment, with its own locality, independence, timing, and selection controls.

All tests should retain negative outcomes, report all declared parameter settings, and separate implementation checks from scientific evidence. None turns a successful hidden-state decoder into a consciousness detector.

12. Discussion: the strongest version of the idea

There is a substantial idea here even if the external world is not an illusion of neural querying. An embodied observer does not first receive a completed world and then think about it. Its evidence is produced by physical interactions with delayed, history-dependent components. Its integrated present is a reconstruction. Its actions help determine what it can distinguish next.

The mathematics gives that picture four concrete footholds. Delay is phase in a narrowband description. Relative phase can control coherent response before detection. State-altering queries have an order-sensitive algebra. The encoder can pull a projective metric back onto the conditions an observer distinguishes. A particular quadratic sampling architecture can even reproduce a Born-shaped formula for internal channels. Those are constructive statements with explicit assumptions, rather than a declaration that all wave observers must see quantum mechanics.

The proposed neural novelty is consequently a candidate *mechanistic connection* between message history, receiver geometry, temporal phase, and accessible reports. State-dependent spikes, travelling waves, predictive equivalence, classical interference, and quantum cognition each have precedents. Combining their vocabularies is not itself a discovery. The work would acquire scientific weight by showing which connection produces a held-out prediction unavailable to simpler models.

The more ambitious conjecture remains open only in the ordinary sense of being incomplete: no account supplied here reproduces the full operational structure of physical quantum theory from neural architecture. Bell correlations under the stated constraints, the universal action scale, external quantization laws, and preparation-and-measurement composition remain independent

obligations. An observer-origin account cannot discharge them by pointing to an introspective resemblance.

If conscious experience belongs to this evolving process, the familiar feeling of one present could reflect a stabilized, predictive integration across many local pasts. That possibility is worth studying without declaring the integration to be consciousness's seat. The useful question becomes: **what properties of an apparent world are consequences of the observer's accessible operations, and which properties remain in public physical records when that observer is replaced?** The state-ping program can investigate the first part. A theory of fundamental physics must answer both.

13. References

- [1] Martin-Burgos, B., Juavinett, A., Riviere, P. D., Hammonds, R., and Voytek, B. (2026). *Action potential waveforms are state-dependent*. bioRxiv preprint. Supplied version posted 21 September 2026; not certified by peer review. doi:10.64898/2026.09.15.751814. Primary source read from the supplied PDF, not independently reproduced.
- [2] Hodgkin, A. L., and Huxley, A. F. (1952). *A quantitative description of membrane current and its application to conduction and excitation in nerve*. Journal of Physiology, 117, 500-544. doi:10.1113/jphysiol.1952.sp004764.
- [3] Zhang, H., Watrous, A. J., Patel, A., and Jacobs, J. (2018). *Theta and alpha oscillations are traveling waves in the human neocortex*. Neuron, 98, 1269-1281.e4. doi:10.1016/j.neuron.2018.05.019.
- [4] Shalizi, C. R., and Crutchfield, J. P. (2001). *Computational mechanics: pattern and prediction, structure and simplicity*. Journal of Statistical Physics, 104, 817-879. arXiv:cond-mat/9907176.
- [5] Wang, Z., Solloway, T., Shiffrin, R. M., and Busemeyer, J. R. (2014). *Context effects produced by question orders reveal quantum nature of human judgments*. Proceedings of the National Academy of Sciences, 111, 9431-9436. doi:10.1073/pnas.1407756111.
- [6] Gabor, D. (1946). *Theory of communication. Part 1: The analysis of information*. Journal of the Institution of Electrical Engineers - Part III: Radio and Communication Engineering, 93, 429-441. doi:10.1049/ji-3-2.1946.0074.
- [7] Everett, H., III (1957). *"Relative state" formulation of quantum mechanics*. Reviews of Modern Physics, 29, 454-462. doi:10.1103/RevModPhys.29.454.
- [8] Zurek, W. H. (2003). *Decoherence, einselection, and the quantum origins of the classical*. Reviews of Modern Physics, 75, 715-775. doi:10.1103/RevModPhys.75.715; arXiv:quant-ph/0105127.
- [9] Bell, J. S. (1964). *On the Einstein Podolsky Rosen paradox*. Physics Physique Fizika, 1, 195-200. doi:10.1103/PhysicsPhysiqueFizika.1.195.
- [10] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). *Proposed experiment to test local hidden-variable theories*. Physical Review Letters, 23, 880-884. doi:10.1103/PhysRevLett.23.880.
- [11] Hensen, B., et al. (2015). *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*. Nature, 526, 682-686. doi:10.1038/nature15759; arXiv:1508.05949.
- [12] Fuchs, C. A., Mermin, N. D., and Schack, R. (2014). *An introduction to QBism with an application to the locality of quantum mechanics*. American Journal of Physics, 82, 749-754. doi:10.1119/1.4874855; arXiv:1311.5253.

[13] Rovelli, C. (1996). *Relational quantum mechanics*. International Journal of Theoretical Physics, 35, 1637-1678. doi:10.1007/BF02302261; arXiv:quant-ph/9609002.

[14] anttiluode (2026). *ChessFlyStatePings*. Repository inspected 2 October 2026 at commit 49385519e55cb7e3b2a06bf3f512ba6b3967bebc. Committed Gate 2 summary. This is an existing artificial-model result, not a new experiment in the present paper.

[15] Provost, J. P., and Vallee, G. (1980). *Riemannian structure on manifolds of quantum states*. Communications in Mathematical Physics, 76, 289-301. doi:10.1007/BF02193559. The metric is a standard construction; its proposed role as an observer-response geometry here is an application, not a novelty claim for the metric itself.

Appendix A. Three short calculations

A.1 Fourier uncertainty with the stated convention

After shifting mean time and frequency to zero, let $\|g\|_2 = 1$. For suitable decay and differentiability, integration by parts gives

$$\operatorname{Re} \int t g(t)^* g'(t) dt = -\frac{1}{2}. \quad (\text{A1})$$

Cauchy-Schwarz therefore implies $\|tg\|_2 \|g'\|_2 \geq 1/2$. Parseval's identity and the transform convention in Equation (31) give $\|tg\|_2 = \Delta t$ and $\|g'\|_2 = \Delta \omega$. This proves Equation (32). For $g(t) \propto \exp[-t^2/(4\sigma_t^2)]$, the intensity variance is σ_t^2 and the spectral variance is $1/(4\sigma_t^2)$, so equality holds. These are widths of the packet's squared magnitudes, not errors of an arbitrary estimator.

A.2 A quadratic-rate race

For independent exponential waiting times with positive total rate $\Lambda = \sum_c \lambda_c$,

$$P(T_b = \min_c T_c) = \int_0^\infty \lambda_b e^{-\Lambda t} dt = \frac{\lambda_b}{\Lambda}. \quad (\text{A2})$$

This proves the first equality in Equation (27). With a finite timeout T , no event has probability $e^{-\Lambda T}$; conditioning on an event before timeout preserves the channel fractions for constant rates. Neither the rate law nor the post-event state update follows from this integral.

A.3 Delay jitter and a three-edge loop

For $\varphi = \bar{\varphi} + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_\varphi^2)$, the characteristic function gives $\mathbb{E}[e^{i\epsilon}] = e^{-\sigma_\varphi^2/2}$. Taking the real part proves Equation (17). For time jitter σ_τ , $\sigma_\varphi = |\Omega| \sigma_\tau$.

As a purely illustrative loop, take three positive-gain edges, each delayed by 5 ms, at 40 Hz. The round-trip delay is 15 ms and $\Phi_C = -2\pi(40)(0.015) = -1.2\pi$, or 0.8π modulo 2π . Local phase redefinitions leave it unchanged. These selected values are not biological parameter estimates.